BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: DIK-SSC-X-2021 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 12 | **Paper Code:** DIK-SSC-X-2021 (A)

Q1. Choose the correct option and fill in empty box, given against each statement:

Q1. For reversible reaction $Kc = [C]^4/[A]^3[B]^2$, the equation will be:

(A) $4C(g) \rightleftharpoons 3A(g) + 2B(g)$
(B) $4C(g) \rightleftharpoons A3(g) + 2B(g)$
(C) $A3(g) + 2B(g) \rightleftharpoons C4(g)$
(D) $A3(g) + 2B(g) \rightleftharpoons 4C(g)$

Q2. Which one of the following is polyprotic acid?

(A) H_3PO_3
(B) NHO_3
(C) CH_3COOH
(D) H_2O

Q3. Which one of the following is a basic salt?

(A) KCl
(B) NaCl
(C) Pb
(D) $KHSO_4$

Q4. Thiophene belongs to the organic compound_____

(A) Htrocyclic
(B) Homocylic
(C) Aromatic
(D) Alicyclic

Q5. In homologous series each member differs from the previous one _____

(A) $-CH-$
(B) $-CH_3-$
(C) $-C_2H-$
(D) $-CH_3-$

Q6. Which one of the following decolourises Br_2 water?

(A) Ethane
(B) Ethene
(C) Propane
(D) Methane

Q7. When ethane reacts with HBr, the compound is formed:

(A) $CH_3 - CH_2 - Br$ (B) $Br - CH_3 - CH_2 - Br$

(C) $CH_3 - CH_2 - CH_3$ (D) Ch = Br

Q8. Fructose (honey) and sucrose are naturally occurring organic compounds?

(A) Vitamins (B) Proteins

(C) Lpids (D) Carbohydrates

Q9. Which of the following is a disaccharide?

(A) Glucose (B) Fructose

(C) Sucrose (D) Starch

Q10. The first layer of atmosphere is _____

(A) Mesosphere (B) Troposphere

(C) Stratosphere (D) Thermosphere

Q11. The salt which causes temporary hardness in water is _____

(A) Magnesium Sulphate (B) Calcium chloride

(C) Magnesium chloride (D) Calcium bicarbonate

Q12. According to the place of occurrence the petroleum exists in solid form is called_____

(A) Asphalt (B) Crude oil

(C) Natural gas (D) Petrol

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 21 | **Paper Code:** DIK-SSC-X-2021 (A)

Q2. Attempt any 8 questions of the following:

(i) Write the equilibrium constant Kc and unit for the following equation. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

(ii) Explain Lewis acid base concept with a proper examples.

(iii) A carbonated drink has $[H^+] = 3.2 \times 10^{-3}$ M classify the drink as neutral, acidic or basic with reason.

(iv) Differentiate between molecular form ula and structural formula with examples for each.

(v) How would you predict the product if $KMnO_4$ solution reacts with alkene?

(vi) Differentiate between simple lipids and complex lipids.

(vii) Mention any four importance of atmosphere.

(viii) Discuss the ozone layer depletion due to: (a) NO_x (b) Atomic oxygen

(ix) Explain hydrogen bonding of water with the help of proper structure.

(x) Make a distinction between soft and hard water.

(xi) Write the two theories about the origin of the petroleum.

SECTION – C

Maximum Marks: 24

> Note: Attempt any 3 questions: Each carry 8 marks.

Q3.

(a) Define equilibrium constant and derive the expression for equilibrium constant for a general reaction.

(b) Write the uses of any three salts along with their names.

Q4.

(a) Define the functional group. Give three example of functional groups containing oxygen.

(b) Write the balance equation for the complete combustion of each of the following: i. Methane ii. Ethene iii. Ethyne

Q5.

(a) Differentiate between glucose and fructose.

(b) Differentiate between mineral, ores and gangue on definition basis.

Q6.

(a) What is acid rain? How acid rain affects building material or statues?

(b) Give the reaction of H_2O with (a). Cl_2 (b). C (c). Si